

5.3 Operating conditions

5.3.1 General operating conditions

Table 24. General operating conditions

Symbol	Parameter	Conditions	Min	Max	Unit
V_{DD}	Standard operating voltage	-	2.0 ⁽¹⁾	3.6	V
V_{IN}	I/O input voltage	-	-0.3	Min ($V_{DD} + 3.6, 5.5$) ⁽²⁾	V
f_{PCLK}	APB clock frequency	-	-	48	MHz
T_A	Ambient temperature ⁽³⁾	Suffix 6 ⁽⁴⁾	-40	85	°C
		Suffix 7 ⁽⁴⁾	-40	105	
		Suffix 3 ⁽⁴⁾	-40	125	
T_J	Junction temperature	Suffix 6 ⁽⁴⁾	-40	105	°C
		Suffix 7 ⁽⁴⁾	-40	125	
		Suffix 3 ⁽⁴⁾	-40	130	

1. When RESET is released functionality is guaranteed down to V_{PDR} min.
2. For operation with voltage higher than $V_{DD} + 0.3$ V, the internal pull-up and pull-down resistors must be disabled.
3. The $T_A(max)$ applies to $P_D(max)$. At $P_D < P_D(max)$ the ambient temperature is allowed to go higher than $T_A(max)$ provided that the junction temperature T_J does not exceed $T_J(max)$. Refer to [Section 6.8: Thermal characteristics](#).
4. Temperature range digit in the order code. See [Section 7: Ordering information](#).

5.3.2 Operating conditions at power-up / power-down

The parameters given in [Table 25](#) are derived from tests performed under the ambient temperature condition summarized in [Table 24](#).

Table 25. Operating conditions at power-up / power-down

Symbol	Parameter	Min	Max	Unit
t_{VDD}	V_{DD} rise time rate	0	∞	$\mu s/V$
	V_{DD} fall time rate	10	∞	

5.3.3 Embedded reset and power control block characteristics

The parameters given in [Table 26](#) are derived from tests performed under the ambient temperature conditions summarized in [Table 24](#).

Table 26. Embedded reset and power control block characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$t_{RSTEMPO}$ ⁽¹⁾	POR temporization when V_{DD} crosses V_{POR}	V_{DD} rising	-	270	500	μs
V_{POR} ⁽¹⁾	Power-on reset threshold	-	1.9	1.94	1.98	V
V_{PDR} ⁽¹⁾	Power-down reset threshold	-	1.88	1.92	1.96	V